

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 93-213

WASTE DISCHARGE REQUIREMENTS
FOR
WOODBIDGE SANITARY DISTRICT
SAN JOAQUIN COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Woodbridge Sanitary District (hereafter Discharger) submitted a Report of Waste Discharge, dated 7 May 1990. The property (Assessor's Parcel No. 015-02-21) is owned by Woodbridge Sanitary District.
2. Waste Discharge Requirements Order No. 73-66, adopted by the Board on 25 August 1972, prescribes requirements for a discharge from Woodbridge Sanitary District to land.
3. Order No. 73-66 is neither adequate nor consistent with current plans and policies of the Board.
4. The Discharger owns and operates a treatment plant that discharges an average of 0.34 million gallons per day (mgd) of domestic waste. The Discharger proposes to discharge an average of 0.5 mgd of treated domestic waste to ponds. The treatment facility consists of two parallel Imhoff tanks having a combined design capacity of 0.5 mgd, two aerated ponds, and five evaporation/percolation ponds, some of which are adjacent to the Mokelumne River.
5. An average influent BOD₅ concentration is 290 mg/l. Approximately 40 percent of the BOD is removed by Imhoff tanks. The aeration ponds reduce it to 35 mg/l before discharging to the evaporation/percolation ponds. Four and one-half days of detention time are provided in each aeration pond. The evaporation/percolation ponds are used alternatively. The facility has three ground water monitoring wells, one upgradient and two downgradient.
6. The treatment plant is in Section 27, T4N, R6E, MDB&M, with surface water drainage to the Mokelumne River, as shown in Attachments A and B, which are attached hereto and part of the Order by reference.
7. The Board adopted a Water Quality Control Plan, Second Edition, for the Sacramento-San Joaquin Delta Basin (5B), which contains water quality objectives for all waters of the Basin. These requirements implement the Basin Plan.

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8. The beneficial uses of the Mokelumne River are municipal, industrial, and agricultural supply; recreation; esthetics enjoyment; navigation; ground water recharge; fresh water replenishment; and preservation and enhancement of fish, wildlife, and other aquatic resources.
9. The beneficial uses of underlying ground water are domestic, industrial, and agricultural supply.
10. The action to update waste discharge requirements for this facility is exempt from the provisions of the California Environmental Quality Act (CEQA) in accordance with Title 14, California Code of Regulations (CCR), Section 15301.
11. This discharge is exempt from the requirements of Title 23, CCR, Section 2510, et seq. (hereafter Chapter 15). The exemption, pursuant to Section 2511(b), is based on the following:
 - a. The Board is issuing waste discharge requirements, and
 - b. The discharge complies with the Basin Plan, and
 - c. The wastewater does not need to be managed according to 22 CCR, Division 4, Chapter 30, as a hazardous waste.
12. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
13. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 73-66 is rescinded and that Woodbridge Sanitary District, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.

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SAN JOAQUIN COUNTY

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3. Discharge of waste classified as 'hazardous' or 'designated', as defined in Sections 2521(a) and 2522(a) of Chapter 15, is prohibited.

B. Discharge Specifications:

1. The monthly average dry weather discharge flow shall not exceed 0.5 million gallons/day.
2. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.
3. Neither the treatment nor the discharger shall cause a nuisance or condition of pollution as defined by the California Water Code, Section 13050.
4. As a means of discerning compliance with Discharge Specification No. 2, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.
5. The treatment facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
6. Ponds shall be managed to prevent breeding of mosquitos. In particular,
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
7. Public contact with wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.
8. Ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration. Freeboard shall never be less than two feet, measured vertically.
9. On or about 1 October of each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification B.8.

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SAN JOAQUIN COUNTY

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C. Sludge Disposal:

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Chapter 15 and approved by the Executive Officer.
2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer at least 90 days in advance of the change.
3. The Discharger is encouraged to comply with the State Guidance Manual issued by the Department of Health Services titled Manual of Good Practice for Landspreading of Sewage Sludge.

D. Ground Water Limitations:

The discharge, in combination with other sources, shall not cause underlying ground water to:

1. Contain waste constituents in concentrations statistically greater than receiving water limits, specified below.
2. Contain chemicals, heavy metals, or trace elements in concentrations that adversely affect beneficial uses or exceed maximum contaminant levels specified in 22 CCR, Division 4, Chapter 15.
3. Exceed a most probable number of total coliform organisms of 2.2/100 ml over any seven-day period.
4. Exceed concentrations of radionuclides specified in 22 CCR, Division 4, Chapter 15.
5. Contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.
6. Contain concentrations of chemical constituents in amounts that adversely affect agricultural use.

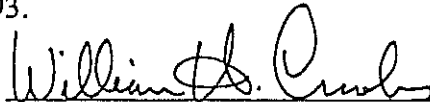
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E. Provisions:

1. The Discharger shall comply with the Monitoring and Reporting Program No. 93-213, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
3. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.
4. At least 90 days prior to termination or expiration of any lease, contract, or agreement involving disposal or reclamation areas or off-site reuse of effluent, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.
5. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
6. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
7. The Board will review this Order periodically and will revise requirements when necessary.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 22 October 1993.


WILLIAM H. CROOKS, Executive Officer

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California Regional Water Quality Control Board

Central Valley Region

Robert Schneider, Chair



Gray Davis
Governor

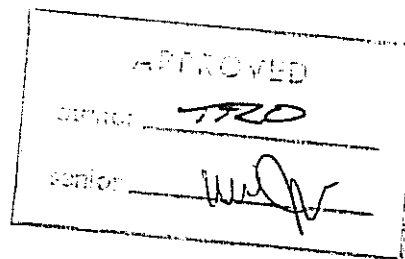
Don H. Hickox
Secretary for
Environmental
Protection

Sacramento Main Office

Internet Address: <http://www.swrcb.ca.gov/rwqcb5>
3443 Routier Road, Suite A, Sacramento, California 95827-3003
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13 September 2001

FILE



Mr. Harold Rohrbach
Woodbridge Sanitary District
19720 N. Benedict Drive
Woodbridge, CA 95258

REVISED MONITORING AND REPORTING PROGRAM NO. 93-213, WOODBRIDGE SANITARY DISTRICT, WOODBRIDGE, SAN JOAQUIN COUNTY

As discussed in our 28 August 2001 meeting, the Regional Water Board is in the process of updating the Waste Discharge Requirements (WDRs) for the Woodbridge Sanitary District. As part of the update process, the Monitoring and Reporting Program (MRP) is being revised. A draft MRP was transmitted to you on 26 June 2001; Woodbridge Sanitary District provided comments on the draft MRP in our meeting on 28 August 2001 and in written correspondence on 20 July 2001.

In our meeting we discussed the condition of Well MW-1 (the northern monitoring well) which historically has contained coliform organisms. You stated that the well may be in need of repair and possibly disinfection. Please evaluate the condition of the well and make needed repairs. If you determine disinfection is required please submit a workplan to accomplish this task prior to performing the work. As we discussed, reliable samples are very important in the Regional Board's evaluation of the wastewater treatment plant and WDRs update process.

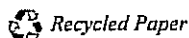
Please telephone Tim O'Brien at (916) 255-3116 if you have any questions regarding the MRP, groundwater monitoring requirements, or the WDR update process.

WENDY WYELS, Chief
Waste Discharge to Land Unit
Lower San Joaquin River Watershed

enc: Revised MRP

cc: Mr. Mike Huggins, San Joaquin County Environmental Health Department, Stockton
Kenneth Kjeldsen, Kjeldsen, Sinnock & Neudeck, Inc., Stockton

California Environmental Protection Agency



CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION
REVISED MONITORING AND REPORTING PROGRAM NO. 93-213

FOR

WOODBIDGE SANITARY DISTRICT
SAN JOAQUIN COUNTY

This revised Monitoring and Reporting Program (MRP) presents requirements for monitoring of the wastewater influent and effluent, treatment/aeration ponds, evaporation/percolation ponds, water supply quality, groundwater, and sludge. This MRP is issued pursuant to Water Code Section 13267. The Discharger shall not implement any changes to this MRP unless and until a revised MRP is issued by the Executive Officer. Specific sample station locations shall be approved by Regional Board staff prior to implementation of sampling activities.

All wastewater samples should be representative of the volume and nature of the discharge. The time, date, and location of each grab sample shall be recorded on the sample chain of custody form. All samples shall be collected and preserved in accordance with EPA and analytical methodology.

INFLUENT MONITORING

Influent samples shall be collected from the sampling station next to the Imhoff tank, and shall be monitored as follows:

<u>Constituents</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow	gallons per day	Metered	Continuous	Monthly
BOD ₅ ¹	mg/l	Grab	Monthly	Monthly
Total Suspended Solids	mg/l	Grab	Monthly	Monthly

¹ Five-day, 20° Celsius Biochemical Oxygen Demand.

EFFLUENT MONITORING

Effluent samples shall be collected from the treatment/aeration pond at the outlet just prior to discharge to the evaporation/percolation ponds. Samples collected from the outlet structure of the pond shall be considered adequately composited. Effluent monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
BOD ₅ ¹	mg/l	Grab	Monthly	Monthly
pH	pH units	Grab	Monthly	Monthly
Nitrate as Nitrogen	mg/l	Grab	Monthly	Monthly
Total Kjeldahl Nitrogen	mg/l	Grab	Monthly	Monthly
Total Dissolved Solids	mg/l	Grab	Monthly	Monthly

¹ Five-day, 20° Celsius Biochemical Oxygen Demand.

POND MONITORING

Each treatment/aeration and evaporation/percolation pond shall be monitored for the following on a weekly basis:

<u>Constituents</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Freeboard	Feet	Measurement	Weekly	Monthly
Dissolved Oxygen	mg/l	Grab	Weekly	Monthly

WATER SUPPLY MONITORING

The quality of the domestic water supply shall be monitored and reported on an annual basis. Submittal of the water quality report from the water provider is acceptable. The monitoring shall include the following:

<u>Constituents</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Standard Minerals ¹	mg/l	Grab	Annually	Annually

¹ Standard Minerals shall include the following compounds: Calcium, Magnesium, Sodium, Chloride, Sulfate, Total Alkalinity (including alkalinity series), and Hardness.

GROUNDWATER MONITORING

There are currently three monitoring wells onsite. Prior to construction of any new groundwater monitoring wells, the Discharger shall submit plans and specifications to the Board for review and approval. Once installed, all new wells shall be added to the MRP, and shall be sampled and analyzed according to the schedule below. All samples shall be collected using approved EPA methods. Water table elevations shall be calculated and used to determine groundwater gradient and direction of flow.

Prior to sampling the groundwater, elevations shall be measured and the wells shall be purged of at least three well volumes until pH and electrical conductivity have stabilized. Depth to groundwater shall be measured to the nearest 0.01 feet. Groundwater monitoring of each well shall include, at a minimum, the following:

<u>Constituents</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sampling/Reporting Frequency</u>
Groundwater Elevation	0.01 feet	Measurement	Quarterly
Nitrate as Nitrogen	mg/l	Grab	Quarterly
Total Kjeldahl Nitrogen	mg/l	Grab	Quarterly
Total Dissolved Solids	mg/l	Grab	Quarterly
Total Coliform Organisms	MPN/100 ml	Grab	Quarterly
pH	pH units	Grab	Quarterly
Standard Minerals ¹	mg/l	Grab	Annually

¹ Standard Minerals shall include the following compounds: Calcium, Magnesium, Sodium, Chloride, Sulfate, Total Alkalinity (including alkalinity series), and Hardness.

SLUDGE

The Discharger shall keep monthly records of the amount of sludge dried in the sludge drying beds, the amount of sludge hauled off site, and the location of sludge disposal. These records shall be submitted with the monthly monitoring reports.

REPORTING

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g., effluent, pond, groundwater well, etc.), and reported analytical result for each sample are readily discernible. The data shall be summarized in such a manner to clearly illustrate compliance with waste discharge requirements and spatial or temporal trends, as applicable. The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported in the next scheduled monitoring report.

As required by the California Business and Professions Code Sections 6735, 7835, and 7835.1, all groundwater monitoring reports shall be prepared under the direct supervision of a California Registered Professional Engineer or Geologist and signed by the registered professional.

A. Monthly Monitoring Reports

Monthly reports shall be submitted to the Regional Board by the 1st day of the second month following the end of the reporting period (i.e. the January monthly report is due by 1 March). At a minimum, the reports shall include:

1. Results of the influent, effluent, and pond monitoring.
2. Results of the sludge monitoring.
3. A discussion of the progress made in acquiring and/or constructing additional storage capacity such that the Discharger is in compliance with its WDRs.
4. A comparison of the monitoring data to the discharge specifications and an explanation of any violation of those requirements. Data shall be presented in tabular format.
5. If requested by staff, copies of laboratory analytical report(s).
6. A calibration log verifying calibration of all monitoring instruments and devices used to fulfill the prescribed monitoring program.

B. Quarterly Monitoring Reports

Quarterly Monitoring Reports shall be submitted to the Regional Board by the 1st day of the second month following the end of the quarter (i.e. the January-March quarter report is due by May 1st) each year. Monthly reports for the months of March, June, September, and December may be submitted as part of the Quarterly Monitoring Report, if desired. The Quarterly Report shall include the following:

1. A narrative description of all preparatory, monitoring, sampling, and analytical testing activities for the groundwater monitoring. The narrative shall be sufficiently detailed to verify compliance with the WDRs, this MRP, and the Standard Provisions and Reporting Requirements. The narrative shall be supported by field logs for each well documenting depth

to groundwater; parameters measured before, during, and after purging; method of purging; calculation of the casing volume; and total volume of water purged.

2. Calculation of groundwater elevations, an assessment of the groundwater flow direction and gradient on the date of measurement, comparison to previous flow direction and gradient data, and discussion of seasonal trends, if any.
3. A narrative discussion of the analytical results for all media and locations monitored, including spatial and temporal trends, with reference to summary data tables, graphs, and appended analytical reports (as applicable).
4. A comparison of monitoring data to groundwater limitations and explanation of any violation of those requirements.
5. Summary data tables of historical and current water table elevations and analytical results.
6. A scaled map showing relevant structures and features of the facility, the locations of monitoring wells and other sampling stations, and groundwater elevation contours referenced to mean sea level datum.
7. Copies of laboratory analytical report(s)

C. Annual Report

An Annual Report shall be prepared as the fourth quarter monitoring report. The Annual Report will include all monitoring data required in the monthly/quarterly schedule. The Annual Report shall be submitted to the Regional Board by 1 February each year. In addition to the data normally presented, the Annual Report shall include the following:

1. The contents of the regular quarterly monitoring report for the last quarter of the year.
2. If requested, tabular and graphical summaries of all data collected during the year;
3. Data for monitoring and analysis performed on an annual basis (i.e., standard minerals);
4. A summary of domestic water quality. The summary may consist of quarterly reports distributed by the domestic water supplier.
5. An evaluation of the performance of the wastewater treatment system, as well as a forecast of the flows anticipated in the next year;
6. An evaluation of the groundwater quality beneath the wastewater treatment facility.
7. A discussion of any data gaps and potential deficiencies/redundancies in the monitoring system or reporting program.
8. A discussion of compliance and corrective action taken, as well as any planned or proposed actions needed to bring the discharge into full compliance with the WDRs.

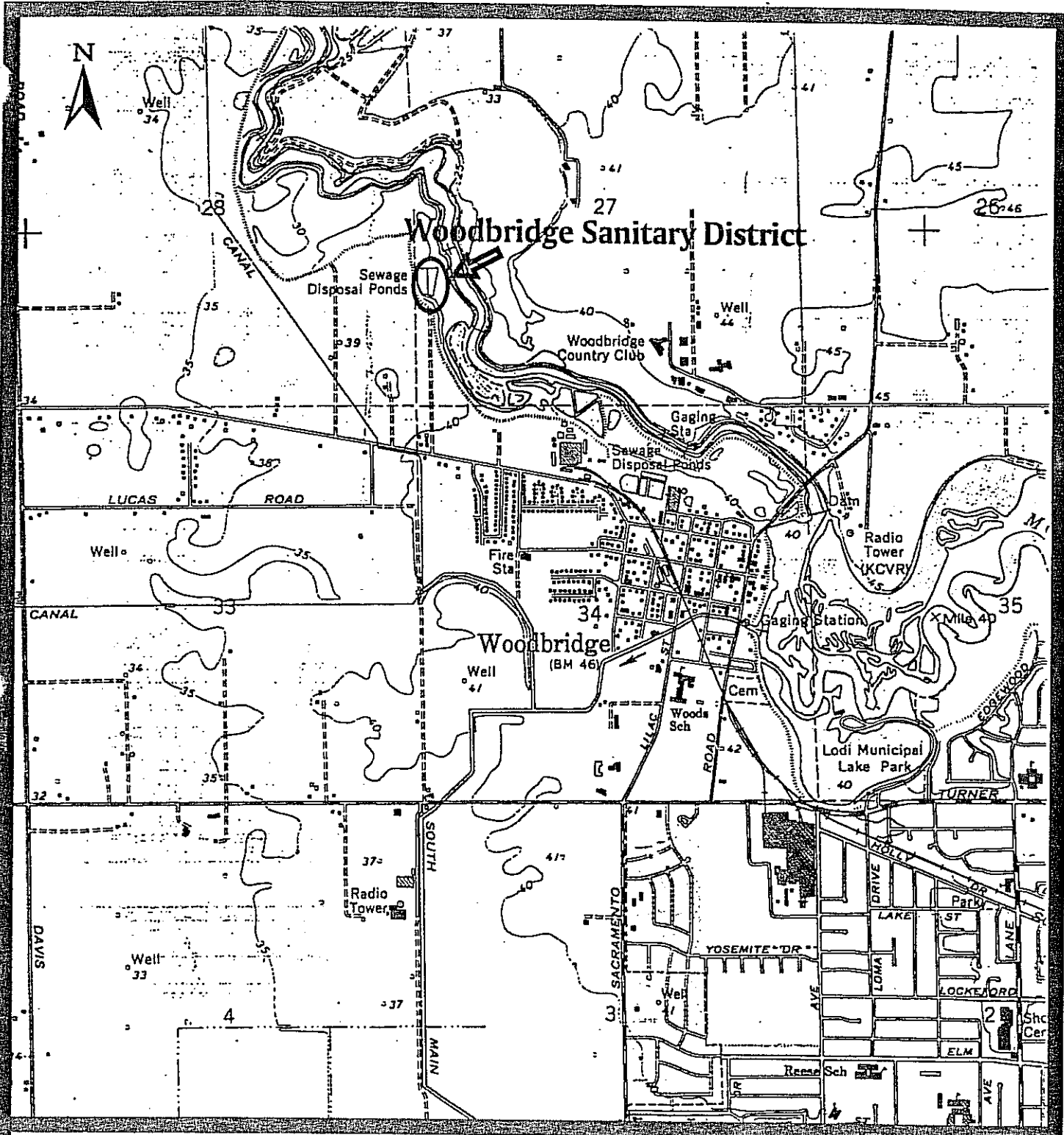
The Discharger shall implement the above monitoring program as of the date of this Order.

for Thomas R Pinkos
GARY M. CARLTON

13 September 2001

TRO: 9/12/2001

Thomas R. Pinkos
Assistant Executive Officer

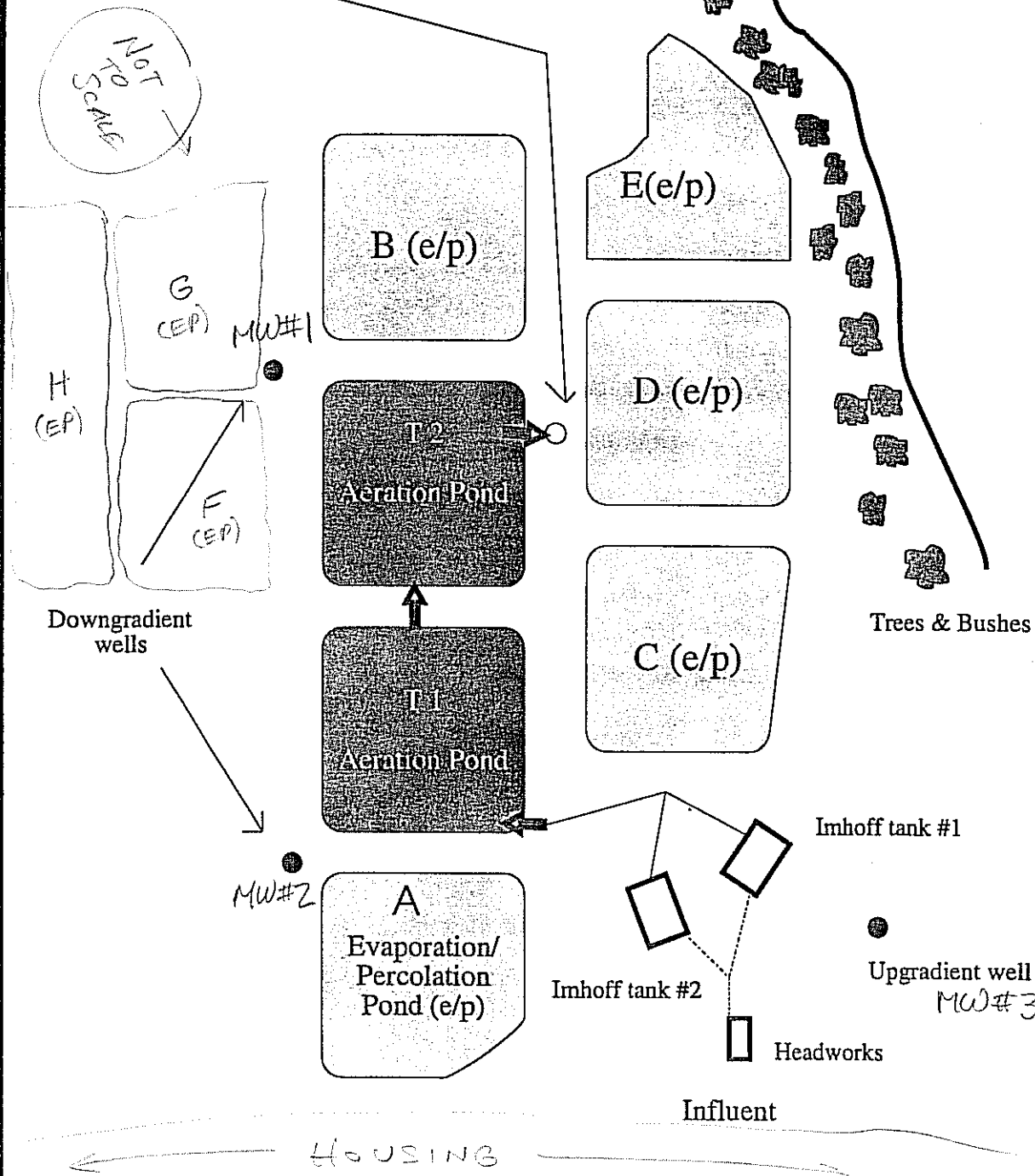


ATTACHMENT A
 WOODBRIDGE SANITARY DISTRICT
 PROJECT LOCATION
 LODI NORTH QUAD., 7.5 MINUTE CALIF.,
 Scale 1 inch = 2000 feet

Mokelumne River



flow split to the various disposal ponds



ATTACHMENT B
WOODBIDGE SANITARY DISTRICT
WASTEWATER TREATMENT SYSTEM
NOT TO SCALE

INFORMATION SHEET

WOODBIDGE SANITARY DISTRICT SAN JOAQUIN COUNTY

Woodbridge Sanitary District is presently regulated by Waste Discharge Requirements Order No. 73-66, which prescribes discharge to land.

Woodbridge Sanitary District owns and operates the facility that currently discharges an average of 0.34 million gallons per day (mgd) of treated domestic wastewater. The design capacity of the plant is 0.50 mgd. The facility treats domestic wastewater in two Imhoff tanks and two aerated ponds, and dispose of it to five evaporation/percolation ponds adjacent to the Mokelumne River.

An average concentration of 290 mg/l (BOD) is received in the influent. Approximately 40 percent of the BOD is removed by the Imhoff tanks. The aeration ponds reduce it to 35 mg/l prior to discharging to the evaporation/percolation ponds. Four and one-half days of detention time are provided in each aeration pond. The aerators operate for about ten hours under current flows and loads. The evaporation/percolation ponds are used alternatively. The facility has three ground water monitoring wells, one upgradient and two downgradient.

Sludge is drawn from the Imhoff tanks monthly. The sludge is dried and composted on-site. Dried sludge has been stored on-site for 12 years and this practice will likely be continued, according to the Mr. George Fiske, plant operator.